PharmaKart

SQL Analysis — Sales & Customer Data

Junior Data Analyst — Take-home assignment

You have joined the Business Operations team as a Junior Data Analyst. The team has sent you a database export and needs help answering a few questions before the next business review.

Suggested time: 2–3 hours **Submission:** SQL file + short findings note

The email from your manager

From: Priya Menon, Business Operations
Subject: Order data — a few numbers for Friday

Hi,

I've attached the latest customer, order and product data.

We're reviewing the business for the last few months and I need some numbers before the meeting on Friday.

Can you spend some time going through the data and help answer a few questions? I want to understand the basic picture first — customers, orders, sales and where the customers are coming from.

After that, please look at the products and coupons. If you notice anything that looks unusual in the data, call it out rather than ignoring it.

One thing to keep in mind: not every order necessarily represents completed business, so think about order status when you calculate sales.

I don't need a fancy dashboard. A clean SQL file and a short summary is enough.

Thanks,
Priya

What you have been given

Table	Rows	What it contains
customers	2,253	Customer details, location, signup/acquisition information
orders	5,147	Orders, dates, status, platform, payment, coupon and amounts
order_items	12,145	Products and quantities included in each order
products	79	Product, category, pricing and product attributes
coupons	10	Coupon codes and campaign/discount information

Part 1 — Get the basic picture

Start with the customer and order data. You decide which queries are needed to answer the manager's questions.

• How many customers, orders and products are in the database?

• How are customers distributed by city/state? Which locations have the most customers?

• How many orders are in each order status?

• What is the total order value and average order value?

• How does order value differ by platform (Android, iOS and Web)?

• Which acquisition channels have brought the most customers?

• Do the acquisition channels also differ when you look at customers who actually placed an order?

Part 2 — What are customers buying?

The sales team wants the analysis to be based on actual order items, not just the product catalogue.

• Which product categories sell the most units?

• Which 10 products have the highest quantity sold?

• Which 10 products generate the most revenue using quantity × unit_price?

• Compare product MRP and selling price. Which products have the largest percentage discount?

• Check whether generic and non-generic products show different sales patterns.

• Check whether prescription and non-prescription products show different sales patterns.

Part 3 — Marketing wants to understand coupons

Marketing has been running coupon campaigns but doesn't know which codes are being used or whether they are reaching customers who actually order.

• Which coupon codes are used most often?

• Join the coupon information to the orders and compare usage across campaign types.

• How much discount has been given through each coupon code?

• Compare customers who used a coupon with customers who did not. Look at order count and spending.

• Find orders with unusually large coupon discounts and investigate which coupon codes are involved.

Part 4 — Finance found a mismatch

Finance noticed that the order-level amounts don't always seem to match what they get when they add up the order items. They asked you to investigate rather than assume one of the reports is wrong.

• Using order_items, calculate the item-level amount as quantity × unit_price.

• Compare the item-level total with the order subtotal for each order.

• Find orders where the two numbers do not match.

• Quantify the difference and check whether coupon discounts, delivery fees or other order-level fields help explain it.

• Document any assumption you make about which order statuses should be included.

Part 5 — Delivery check

The operations team has also asked for a quick look at delivery performance.

• Calculate delivery time from order_date to delivery_date where both dates are available.

• Find the average delivery time.

• Compare delivery time by platform and by location.

• Identify orders with unusually long delivery times. Explain how you decided what counts as unusual.

Part 6 — Build one customer report

Management wants one row per customer so they can use the result later. Create a query that produces a customer-level report containing, where available:

• customer_id

• customer name

• city/state

• acquisition channel

• total orders

• total spend

• average order value

• last order date

• coupon usage

• customer segment

Create a simple customer segment using total spending. You can choose the thresholds, but explain why you chose them.

Final question

You have 30 minutes before the management meeting. Write three observations from the data. For each one, give the number that supports it and explain why the observation might matter to the business. Do not claim a cause unless the data actually supports it.

Submission

• analysis.sql — your complete SQL work

• findings.pdf or findings.docx — maximum 2 pages

• README.md — assumptions, database setup and any decisions you made

There is no single correct SQL style. What matters is that your result is correct, readable and that you can explain how you got there.